SnapFix

AI-assisted Home Repair with Manual Evidence

Photo diagnosis, safety decisions, repair tracking and bilingual reporting

COMP7506 Group Project

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Abstract

SnapFix is a smart-home maintenance application for tenants and household users who can observe a problem but may not know the appliance model, relevant instructions, or safest next step. The application combines Qwen3-VL visual extraction, BM25 manual retrieval, deterministic safety rules, chunk-level citation validation, persistent repair progress, and editable Chinese/English repair requests. Rather than treating a model response as authoritative, SnapFix separates visible observations from possible causes and withholds unsupported do-it-yourself instructions. The current implementation includes real accounts, source-linked guidance, safety refusal, and a repair archive. The 14 November 2026 review passed 81 backend tests and checked 26 controlled image-quality variants. Earlier repository summaries report five Flutter tests and selected real Qwen workflows; those were not rerun in this review. These checks demonstrate implementation and workflow reliability within their scope, not universal diagnostic accuracy.

Keywords: smart home, appliance repair, multimodal AI, retrieval-augmented generation, safety, mobile application

1 Background Research

1.1 Market context

Household repair guidance is distributed across repair communities, visual search tools, and professional consultation services. Each addresses a different part of the user journey, but a tenant still needs to connect device identity, applicable instructions, safety, and communication with a landlord or technician. Our comparison is based on published product documentation, not a controlled hands-on reliability comparison.

Application	Documented capabilities	Design question relevant to SnapFix
iFixit / FixBot	Step-by-step repair guides, a repair knowledge base, and visual assistance through questions and photos. [1]	Users still need to confirm that guidance applies to their exact device. SnapFix keeps model identity, uncertainty, and evidence visible.
Frontdoor	Video consultations with repair experts, saved call notes, and referrals to local professionals. [2]	Live consultation depends on connectivity, expert availability, and regional coverage. SnapFix prepares a concise bilingual request before contact.
Google Lens	Visual search for objects, similar images, related websites, and AI overviews. [3]	Visual similarity alone does not establish a model-specific fault or validated procedure. SnapFix uses visual extraction as an input to retrieval, not as sufficient repair evidence.

The open-source FixIt Lens project is an additional engineering reference for image input, retrieval, safety checks, and evaluation; it is not counted as one of the three market applications above. [4]

1.2 Opportunity

The opportunity is the integration of a narrow rental workflow rather than a claim that any individual capability is unique. SnapFix collects device information, presents supporting instructions, decides whether professional help is appropriate, and prepares a repair request in Chinese or English. The intended user value is a clear and traceable next step, especially when a tenant cannot safely or confidently perform a repair.

2 Application Overview

2.1 Category and motivation

SnapFix is an **information and household maintenance application**. Its target users are students and tenants who encounter unfamiliar appliances, missing manuals, or error codes. Three recurring problems motivate the design:

1. A user can see a warning light or error code but cannot reliably identify the applicable instruction.
2. A user may not know whether a task is non-destructive, model-specific, or dangerous.
3. After deciding to seek help, the user still needs to describe the device, observed issue, attempted steps, and safety concerns clearly.

The product principle is therefore **"Snap it. Fix it. Or report it."** Diagnosis is not treated as complete until the user receives either evidence-linked guidance or a safe route to professional help.

2.2 Design and features

The mobile flow supports camera or library input, a manual error-code path, analysis progress, a diagnosis report, step feedback, repair status, an archive, and bilingual repair drafts. The interface supports English and Simplified Chinese, light/dark/system appearance, and reduced motion. Real accounts support email, Sign in with Apple, and Google OAuth, with per-user isolation of sessions and preferences.

Feature	Delivered behavior
Photo diagnosis	Extracts device category, brand, model, error code, visible symptoms, and confidence without requiring unknown fields.
Manual lookup	Accepts brand and error-code input without requiring a camera.
Grounded steps	Displays exact manual chunks, original excerpts, PDF pages, and source URLs.
Safety guardrails	Applies deterministic rules before generation and again to generated steps.
Repair progress	Persists done, skipped, and not-useful feedback; repeated taps undo feedback.
Repair status	Supports investigating, awaiting-repair, and resolved states with archive filters.
Bilingual reports	Allows editing, saving, resetting, copying, and sharing Chinese or English drafts.
Explicit demo mode	Uses labelled deterministic scenarios without model credentials; unknown images are rejected.

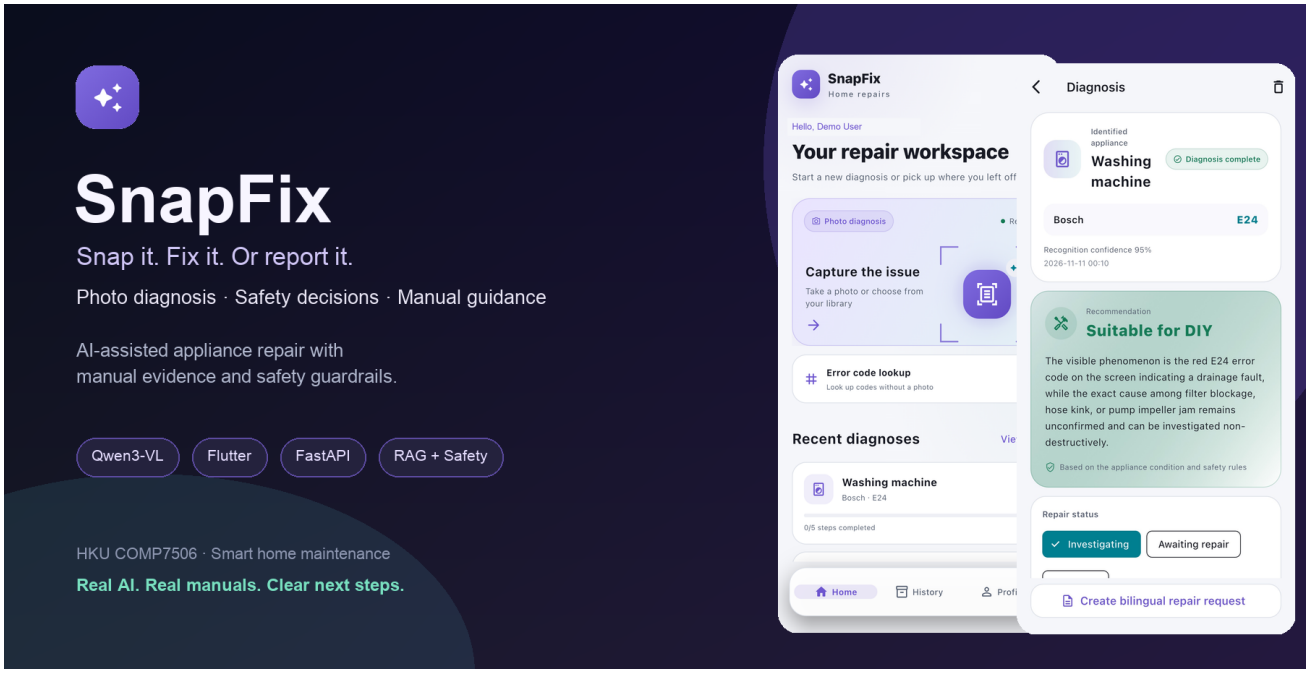


Figure 1: Earlier application screens illustrating the interface. The pictured washer E24 result is obsolete; the current mock is a dishwasher with professional referral and no repair steps. Personal information is replaced by a demo label.

3 System and AI Design

3.1 Architecture

SnapFix uses a Flutter application and an authenticated FastAPI backend. Qwen3-VL is the live provider described by the integration through an OpenAI-compatible API; Gemini remains an optional provider. SQLite stores user accounts, token hashes, diagnoses, step feedback, repair status, report drafts, and the imported manual corpus. Original photographs are not written to the database.

3.2 Diagnosis pipeline

1. **Image quality.** Validate file size, dimensions, brightness, sharpness, and readability before model invocation.
2. **Vision extraction.** Produce structured JSON for device identity and visible observations. The prompt prohibits invented models, error codes, or internal faults.
3. **Safety gate.** Refuse high-risk categories before generation, including mains electricity, gas, refrigerant, lithium batteries, pressure equipment, and microwave components.
4. **BM25 retrieval.** Rank real manual chunks using category filters, brand aliases, bilingual expansion, and normalized error codes.

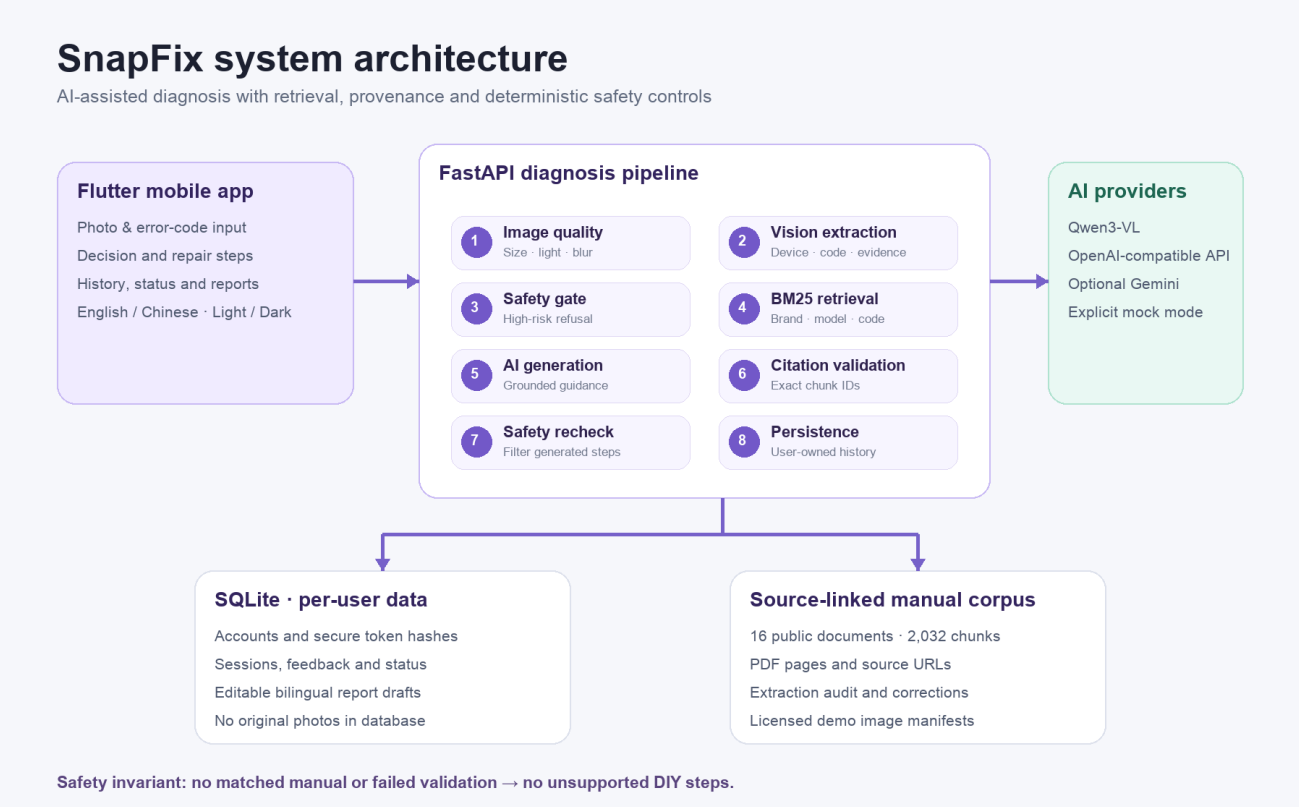


Figure 2: System architecture and the separation of AI generation from deterministic safety and evidence validation.

5. **Grounded generation.** Request a decision and steps constrained to retrieved evidence and the current interface language.
6. **Citation validation.** Accept only exact chunk IDs from the retrieval result and preserve original text and PDF pages.
7. **Generated-content safety review.** Drop generated steps that introduce dangerous operations even when a citation is present.
8. **Persistence.** Save the resulting session and subsequent repair activity under the authenticated user.

This design deliberately separates probabilistic model output from deterministic safety and evidence checks. Low-confidence recognition, missing manual coverage, invalid citations, and unsafe generated content all lead to an explicit non-DIY outcome. A live provider failure returns an error and never substitutes an unrelated mock diagnosis.

4 Implementation and Evaluation

4.1 Corpus and provenance

The current corpus contains 16 public documents and 2,032 text chunks. Records include source URLs, PDF pages, stable chunk IDs, source hashes, and extraction metadata. A full rebuild from the bundled PDFs on 14 November 2026 reproduced the committed chunks, manual records, and extraction audit byte for byte. Of 599 audited pages, 548 have extracted text, 45 are sparse, and six are excluded as undecodable; one page has a source-bound visual correction. These counts describe extraction coverage, not expert validation of every repair procedure. Public demonstration photographs include author, licence, hash, and usage metadata. Appearance images are used for workflow and image-quality testing, not as confirmed fault examples.

4.2 Current verification

Verification	Current result	Scope
Backend regression	81 passed	Reproduced 14 November 2026: baseline 46 plus 35 mock and data acceptance checks
Flutter tests	Previously: 5 passed	Prior result; two new error-flow tests added but not run (SDK unavailable)
Flutter analysis	Previously: no issues	Repository-reported static checks; not rerun
Image-quality variants	26 controlled cases	Dark/blur retake behavior, not diagnostic accuracy
Real Qwen HTTP flows	Prior summary only	Router, microwave, Bosch E24; per-run records not bundled, not rerun
Manual corpus	16 documents / 2,032 chunks	Available evidence coverage, not universal appliance support

The backend suite uses an isolated temporary database and does not call a live model. It now verifies S1–S6, alternative inputs, asset integrity, correct Bosch E24 mock routing, unknown-image rejection, and safety enforcement in mock mode. The E24 mock is a dishwasher with an unknown model and professional referral, not evidence of a confirmed fault. Earlier live-provider summaries require a recording commit, input, output, citations, and timing before live acceptance is complete. The existing 128.633-second video decodes fully but has silent audio; narration, subtitles, and missing evidence interactions remain to be completed.

4.3 Limitations

The current build is a course project and local development deployment, not a public repair service. The corpus covers a limited set of devices, and the present evaluation does not establish broad fault-recognition accuracy. Camera permissions, notifications, system sharing, and OAuth still require physical-device validation. Future work should expand confirmed-fault photographs, improve manual coverage, evaluate a larger labelled dataset, and add production monitoring and public privacy documentation without weakening citation or safety requirements.

5 Contributions

The integrated team mainline combines selected outputs from the original team workstreams. The table records the original team roles and the sources referenced by the integrated repository; it does not imply that every team feature was merged unchanged.

Member	Original role	Workstream and coordination responsibilities
Cao Junjia	Team lead; BE-1 pipeline	Team planning and task coordination; backend architecture, API contracts, providers and safety; integration review, PR coordination, and submission planning
Shi Junying	BE-2 retrieval and storage	BM25 retrieval, SQLite persistence, reporting endpoints, and backend documentation
Zhu Yutaimin	FE-1 scan and diagnosis	Flutter foundations, shared components, home, camera, preview, and diagnosis flow
Yang Ruijia	FE-2 guidance and archive	Step guidance, repair history, report interactions, visual design, and video materials
Zhang Fan	Data and content	Corpus ETL, demonstration scenarios, market research, acceptance records, and attributed image assets

For the data and content workstream, the current artifact review and outstanding acceptance work are recorded in `docs/zhang-fan-review.md`. It maps corpus ETL, demo scenarios, market research, acceptance evidence, and attributed images to repository files. This artifact-level check does not establish personal authorship from the role table alone.

Each member should confirm the final description and add work completed outside GitHub before submission. Each member of a multi-person team must also submit the course Peer Review Form separately.

References

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[4] FixIt Lens repository, engineering reference. https://github.com/rakexhs/fixit-lens. Reviewed 30 October 2026; source rechecked 14 November 2026.